

Amazon Sale Report EDA Project

Aim of the Project

The aim of this project is to perform Exploratory Data Analysis (EDA) on Amazon sale report data to:

- Understand the overall structure, patterns, and trends in the sales.
- Identify and clean missing or duplicate data.
- Detect outliers and analyze their effect on sales insights.
- Visualize sales performance by category, location, fulfillment method, and over time.
- Reveal relationships between variables using correlation analysis.

✓ *Importing the modules*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

✓ *Loading the data*

```
df = pd.read_csv("Amazon Sale Report.csv")
df
```

	index	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	...	currency	Amount
0	0	405-8078784-5731545	04-30-22	Cancelled	Merchant	Amazon.in	Standard	SET389	SET389-KR-NP-S	Set	...	INR	647.60
1	1	171-9198151-1101146	04-30-22	Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard	JNE3781	JNE3781-KR-XXXL	kurta	...	INR	406.00
2	2	404-0687676-7273146	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3371	JNE3371-KR-XL	kurta	...	INR	329.00
3	3	403-9615377-8133951	04-30-22	Cancelled	Merchant	Amazon.in	Standard	J0341	J0341-DR-L	Western Dress	...	INR	753.30
4	4	407-1069790-7240320	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3671	JNE3671-TU-XXXL	Top	...	INR	574.00
...	...	...	...	...	...	...	...	...	...	...	...	...	...
128970	128970	406-6001380-7673107	05-31-22	Shipped	Amazon	Amazon.in	Expedited	JNE3697	JNE3697-KR-XL	kurta	...	INR	517.00
128971	128971	402-9551604-7544318	05-31-22	Shipped	Amazon	Amazon.in	Expedited	SET401	SET401-KR-NP-M	Set	...	INR	999.00
128972	128972	407-9547469-3152358	05-31-22	Shipped	Amazon	Amazon.in	Expedited	J0157	J0157-DR-XXL	Western Dress	...	INR	690.00
128973	128973	402-6184140-0545956	05-31-22	Shipped	Amazon	Amazon.in	Expedited	J0012	J0012-SKD-XS	Set	...	INR	1199.00
128974	128974	408-7436540-8728312	05-31-22	Shipped	Amazon	Amazon.in	Expedited	J0003	J0003-SET-S	Set	...	INR	696.00

128975 rows × 24 columns

Shape of the dataset

```
df.shape
```

```
(128975, 24)
```

Extracting of the data

```
df.head()
```

	index	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	...	currency	Amount	
0	0	405-8078784-5731545	04-30-22	Cancelled	Merchant	Amazon.in	Standard	SET389	SET389-KR-NP-S	Set	...	INR	647.62	
1	1	171-9198151-1101146	04-30-22	Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard	JNE3781	JNE3781-KR-XXXL	kurta	...	INR	406.00	BE
2	2	404-0687676-7273146	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3371	JNE3371-KR-XL	kurta	...	INR	329.00	NAV
3	3	403-9615377-8133951	04-30-22	Cancelled	Merchant	Amazon.in	Standard	J0341	J0341-DR-L	Western Dress	...	INR	753.33	PUD
4	4	407-1069790-7240320	04-30-22	Shipped	Amazon	Amazon.in	Expedited	JNE3671	JNE3671-TU-XXXL	Top	...	INR	574.00	

5 rows × 24 columns

df.tail()

	index	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	...	currency	Amount	
128970	128970	406-6001380-7673107	05-31-22	Shipped	Amazon	Amazon.in	Expedited	JNE3697	JNE3697-KR-XL	kurta	...	INR	517.0	
128971	128971	402-9551604-7544318	05-31-22	Shipped	Amazon	Amazon.in	Expedited	SET401	SET401-KR-NP-M	Set	...	INR	999.0	
128972	128972	407-9547469-3152358	05-31-22	Shipped	Amazon	Amazon.in	Expedited	J0157	J0157-DR-XXL	Western Dress	...	INR	690.0	
128973	128973	402-6184140-0545956	05-31-22	Shipped	Amazon	Amazon.in	Expedited	J0012	J0012-SKD-XS	Set	...	INR	1199.0	
128974	128974	408-7436540-8728312	05-31-22	Shipped	Amazon	Amazon.in	Expedited	J0003	J0003-SET-S	Set	...	INR	696.0	

5 rows × 24 columns

▼ Info Of The DataSet

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 128975 entries, 0 to 128974
Data columns (total 24 columns):
#   Column                Non-Null Count  Dtype
---  -
0   index                 128975 non-null  int64
1   Order ID              128975 non-null  object
2   Date                  128975 non-null  object
3   Status                128975 non-null  object
4   Fulfilment            128975 non-null  object
5   Sales Channel         128975 non-null  object
6   ship-service-level    128975 non-null  object
7   Style                 128975 non-null  object
8   SKU                   128975 non-null  object
9   Category              128975 non-null  object
10  Size                  128975 non-null  object
```

```

11 ASIN                128975 non-null object
12 Courier Status      122103 non-null object
13 Qty                128975 non-null int64
14 currency           121180 non-null object
15 Amount             121180 non-null float64
16 ship-city          128942 non-null object
17 ship-state         128942 non-null object
18 ship-postal-code   128942 non-null float64
19 ship-country       128942 non-null object
20 promotion-ids      79822 non-null object
21 B2B                128975 non-null bool
22 fulfilled-by       39277 non-null object
23 Unnamed: 22        79925 non-null object
dtypes: bool(1), float64(2), int64(2), object(19)
memory usage: 22.8+ MB

```

▼ DESCRIPTIVE STATISTICS

Descriptive Statistics (Numerical)

```
df.describe()
```

	index	Qty	Amount	ship-postal-code
count	128975.000000	128975.000000	121180.000000	128942.000000
mean	64487.000000	0.904431	648.561465	463966.236509
std	37232.019822	0.313354	281.211687	191476.764941
min	0.000000	0.000000	0.000000	110001.000000
25%	32243.500000	1.000000	449.000000	382421.000000
50%	64487.000000	1.000000	605.000000	500033.000000
75%	96730.500000	1.000000	788.000000	600024.000000
max	128974.000000	15.000000	5584.000000	989898.000000

Descriptive Statistics (Categorical)

```
df.describe(include='object')
```

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	Size	ASIN	Courier Status
count	128975	128975	128975	128975	128975	128975	128975	128975	128975	128975	128975	122103
unique	120378	91	13	2	2	2	1377	7195	9	11	7190	5
top	403-4984515-8861958	05-03-22	Shipped	Amazon	Amazon.in	Expedited	JNE3797	JNE3797-KR-L	Set	M	B09SDXFFQ1	Shipped
freq	12	2085	77804	89698	128851	88615	4224	773	50284	22711	773	109487

▼ Dropping unnecessary columns

```
df = df.drop(columns=['Unnamed: 22', 'index'], errors='ignore')
```

```
df.columns
```

```

Index(['Order ID', 'Date', 'Status', 'Fulfilment', 'Sales Channel ',
      'ship-service-level', 'Style', 'SKU', 'Category', 'Size', 'ASIN',
      'Courier Status', 'Qty', 'currency', 'Amount', 'ship-city',
      'ship-state', 'ship-postal-code', 'ship-country', 'promotion-ids',
      'B2B', 'fulfilled-by'],
      dtype='object')

```

▼ Duplicates check and Removal

```
df[["SKU"]]
```

	SKU
0	SET389-KR-NP-S
1	JNE3781-KR-XXXL
2	JNE3371-KR-XL
3	J0341-DR-L
4	JNE3671-TU-XXXL
...	...
128970	JNE3697-KR-XL
128971	SET401-KR-NP-M
128972	J0157-DR-XXL
128973	J0012-SKD-XS
128974	J0003-SET-S

128975 rows × 1 columns

What is SKU?

A *SKU*, or *Stock Keeping Unit*, is a unique alphanumeric code that businesses use to identify and track their products internally

Purpose:

SKUs are used to differentiate between products, especially when variations like color, size, or style exist. For example, a t-shirt might have one SKU for a small, blue version and another for a large, red version.

Source - **GOOGLE**

```
df.drop_duplicates(['Order ID', 'SKU', 'Date'],keep = 'first', inplace = True)
```

```
print("After dropping duplicates shape of the dataset is\n")
df.shape
```

After dropping duplicates shape of the dataset is

(128968, 22)

Null values counts

```
df.isnull()
```

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	Size	...	Qty	currency	Amount	ship-city
0	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
128970	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
128971	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
128972	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
128973	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False
128974	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False

128968 rows × 22 columns

```
df.isnull().sum()
```

	0
Order ID	0
Date	0
Status	0
Fulfilment	0
Sales Channel	0
ship-service-level	0
Style	0
SKU	0
Category	0
Size	0
ASIN	0
Courier Status	6872
Qty	0
currency	7792
Amount	7792
ship-city	33
ship-state	33
ship-postal-code	33
ship-country	33
promotion-ids	49150
B2B	0
fulfilled-by	89691

dtype: int64

No of Unique values in each Column

```
print("\n--- Unique Values Per Column ---")
print(df.nunique())
```

```
--- Unique Values Per Column ---
Order ID      120378
Date           91
Status        13
Fulfilment     2
Sales Channel  2
ship-service-level  2
Style        1377
SKU          7195
Category       9
Size          11
ASIN         7190
Courier Status  3
Qty           10
currency       1
Amount        1410
ship-city     8955
ship-state     69
ship-postal-code 9459
ship-country   1
promotion-ids  5787
B2B            2
fulfilled-by   1
dtype: int64
```

```
df.dropna(inplace=True)
```

```
df.isnull().sum()
```

	0
Order ID	0
Date	0
Status	0
Fulfilment	0
Sales Channel	0
ship-service-level	0
Style	0
SKU	0
Category	0
Size	0
ASIN	0
Courier Status	0
Qty	0
currency	0
Amount	0
ship-city	0
ship-state	0
ship-postal-code	0
ship-country	0
promotion-ids	0
B2B	0
fulfilled-by	0

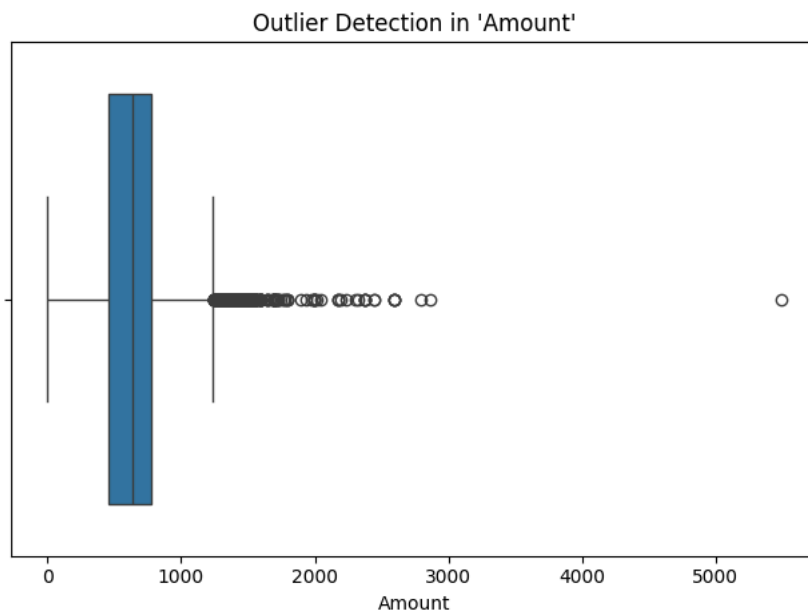
dtype: int64

df.shape

(32338, 22)

OUTLIER DETECTION (Boxplot for Amount)

```
plt.figure()
sns.boxplot(x=df['Amount'])
plt.title("Outlier Detection in 'Amount'")
plt.tight_layout()
plt.show()
```

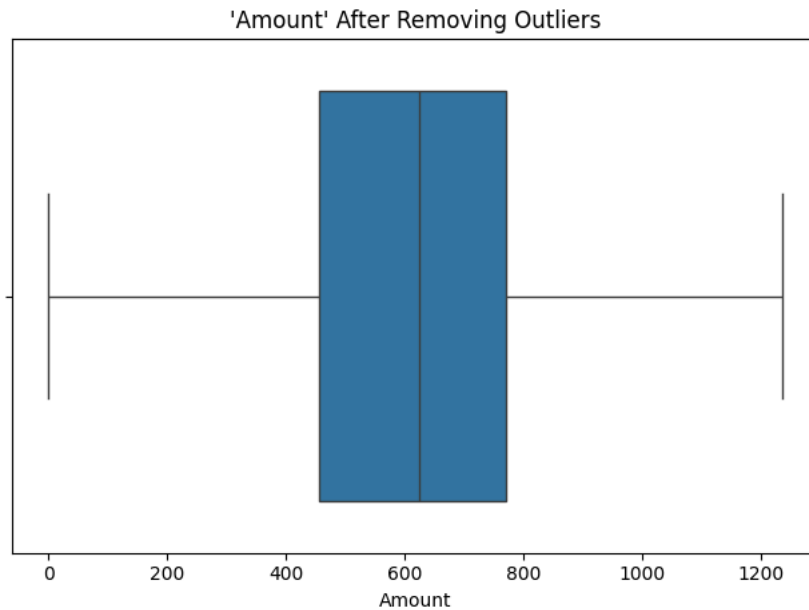


Create a filtered copy for outlier-removed analysis (IQR method)

```
Q1 = df['Amount'].quantile(0.25)
Q3 = df['Amount'].quantile(0.75)
IQR = Q3 - Q1
filtered_df = df[(df['Amount'] >= Q1 - 1.5 * IQR) & (df['Amount'] <= Q3 + 1.5 * IQR)]
```

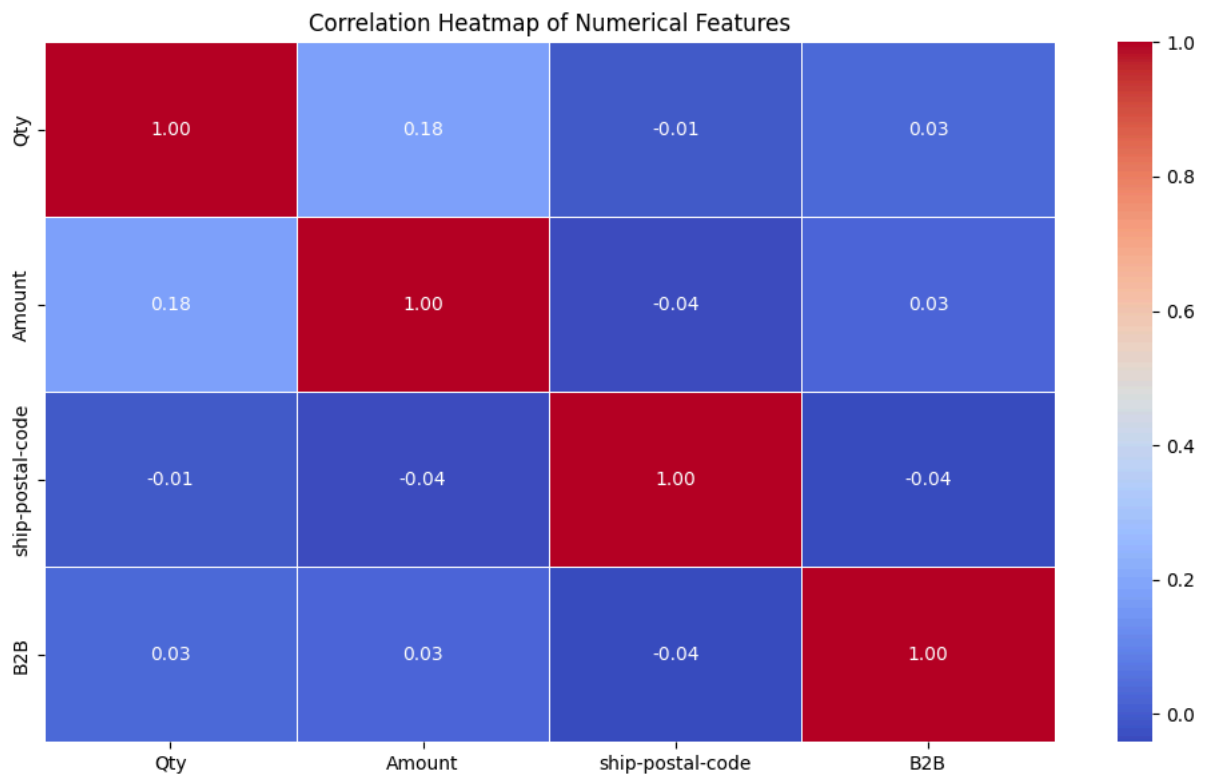
✓ *boxplot after removing outliers*

```
plt.figure()
sns.boxplot(x=filtered_df['Amount'])
plt.title("'Amount' After Removing Outliers")
plt.tight_layout()
plt.show()
```



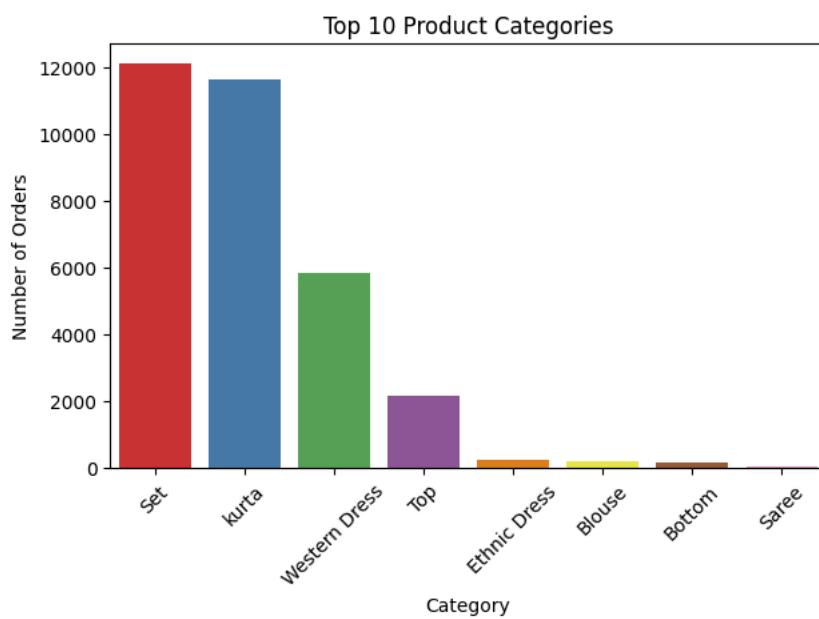
✓ *Correlation Heatmap*

```
plt.figure(figsize=(10, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', linewidths=0.5, fmt=".2f")
plt.title("Correlation Heatmap of Numerical Features")
plt.tight_layout()
plt.show()
```

✓ **Top Product Categories**

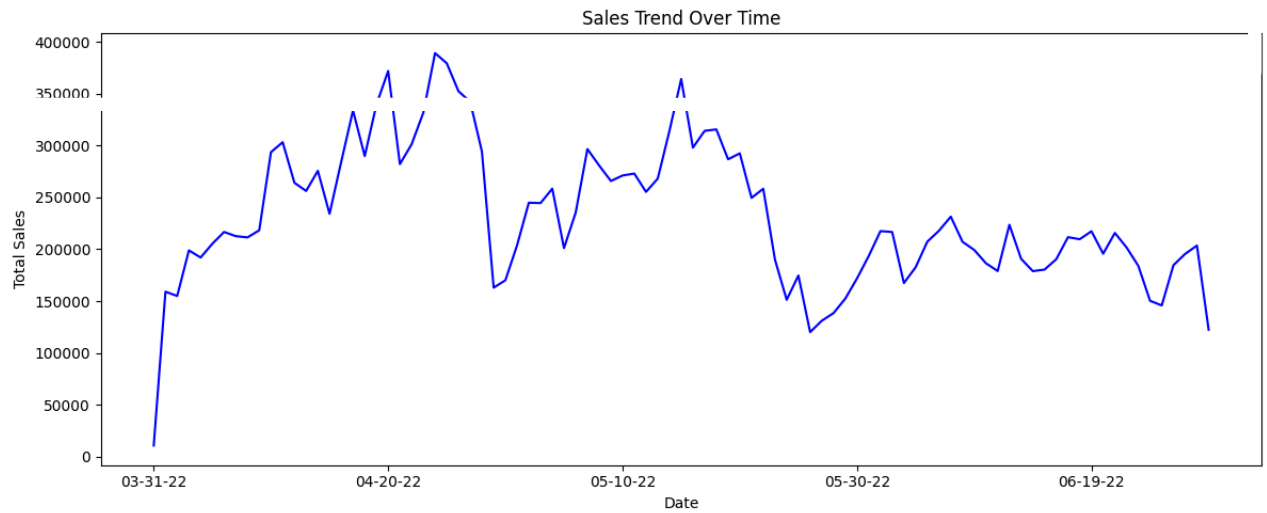
```
top_categories = df['Category'].value_counts().head(10)
plt.figure()
sns.barplot(x=top_categories.index, y=top_categories.values, palette="Set1")
plt.title("Top 10 Product Categories")
plt.xticks(rotation=45)
plt.ylabel("Number of Orders")
plt.xlabel("Category")
plt.tight_layout()
plt.show()
```



✓ **SALES TREND OVER TIME**

```
df_grouped_by_date = df.groupby('Date')['Amount'].sum()
plt.figure(figsize=(12,5))
df_grouped_by_date.plot(color='blue')
plt.title('Sales Trend Over Time')
```

```
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.tight_layout()
```



Top Shipping Cities

```
top_cities = df['ship-city'].value_counts().head(10)
plt.figure()
sns.barplot(x=top_cities.values, y=top_cities.index, color="red", edgecolor="black")
plt.title("Top 10 Shipping Cities")
plt.xlabel("Number of Orders")
plt.ylabel("City")
plt.tight_layout()
plt.show()
```

